

JINGYU LIU

✉ jin.liu.0201@gmail.com · ☎ (+86) 159-1532-4985 · 🌐 [Vieloooo](#)

EDUCATION

Hong Kong University of Science and Technology, Guangzhou, China **09/2024 – 06/2028**
(expected)

PhD in Financial Technology. Advised by Prof. Xuechao Wang and Prof. Yingjie Xue

Harbin Institute of Technology, Shenzhen, China **09/2019 – 06/2023**

B.E. in Computer Science and Technology

PUBLICATIONS/PREPRINTS

Jingyu Liu, Yingjie Xue, Di Wu, Jian Liu, Xuechao Wang. **Zeus: Defending against Fee Stealing and Griefing Attacks in Multi-Hop Payments**. IACR, 2025.

Zifan Peng, Yingjie Xue, **Jingyu Liu**. **Cross-Chain Options: A Bridgeless, Universal, and Efficient Approach**. In the 2025 IEEE International Conference on Web Services (ICWS'25), July 7-12, 2025, Helsinki, Finland.

Jingyu Liu, Yingjie Xue, Zifan Peng, Chao Lin, Xinyi Huang. **FairRelay: Fair and Cost-Efficient Peer-to-Peer Content Delivery through Payment Channel Networks**. Arxiv, 2024.

Jingyu Liu, Yang Zhao, Xinrui Xiao, and Jie Liu. 2023. **CAS: Crop Aerial Sensing Simulation in Smart Farming** In the 2023 20th Annual IEEE International Conference on Sensing, Communication, and Networking (SECON'23), Workshop on Sensing, Communication and Networking for Smart Agriculture, September 11–14, 2023, Madrid, Spain.

Jingyu Liu, Xinrui Xiao, Yang Zhao, and Jie Liu. 2022. **Demo Abstract: Containerized Mobile Sensing Simulation Framework for Smart Agriculture**. In the 20th ACM Conference on Embedded Networked Sensor Systems (SenSys'22), November 6–9, 2022, Boston, MA, USA. ACM, New York, NY, USA.

RESEARCH AND WORK EXPERIENCE

Thrust of Financial Technology

Hong Kong University of Science and Technology, Guangzhou, China

12/2023 – 08/2024

Full-time Research Assistant | Advisor: [Prof. Yingjie Xue](#)

- Proposed FairRelay, the first cost-efficient decentralized content delivery protocol with fair incentives over Payment Channel Networks, utilizing multiple cryptographic primitives such as zero-knowledge proofs, homomorphic encryption, and fraud-proof schemes.
- Formally proved the security properties of FairRelay within the *Universal Composability Framework* and thoroughly evaluated its efficiency both on-chain and off-chain.

International Research Institute for Artificial Intelligence

Harbin Institute of Technology, Shenzhen, China

08/2022 – 06/2023

Part-time Undergraduate Research Assistant | Advisor: [Prof. Yang Zhao](#)

- Developed a containerized mobile sensing simulation (CMOS) [framework](#) for smart agriculture applications. The CMOS framework is an out-of-the-box solution for deploying unmanned vehicle simulations and generating synthetic datasets specific to the agriculture domain. Implemented using ROS (Robot Operating System), integrating Blender, Gazebo, MAVLink, and PX4 autopilots in Docker.
- Generated a large-scale synthetic image dataset of cornfields based on the aforementioned framework, used for training deep learning models on the corn leaf image segmentation problem. Our segmentation model outperformed the state-of-the-art segmentation model SAM with higher efficiency in corn leaf segmentation. This work resulted in a conference paper presented at SECON'23.

Software and Emerging Technologies Lab

Polytechnique Montréal, Montreal, Canada

07/2022 – 10/2022

Full-time Research Intern | Advisor: [Prof. Mohammad Hamdaqa](#)

- Proposed a multi-level [ontology](#) of residential lease agreements by modeling standard residential lease forms provided across Canada using Protégé, employing Giancarlo Guizzardi's [approach](#).
- Based on the ontology, developed a set of domain-specific language (DSL) [tools](#) on JetBrains MPS, including two domain-specific languages: AliceGen and AliceVerify, which were used to formalize and verify leases written in natural language, subsequently converting them into Smart Contracts.

SELECTED PROJECTS

[Atomic Buy](#)

A Decentralized Framework for Trustless and Atomic Digital Content Purchases over the Lightning Network

[RBTC](#)

Bitcoin-like SegWit UTXO blockchain node and cli-wallet in Rust, implementing P2WPKH payment.

MISCELLANEOUS

- Welcome to my Blog: <https://vieloooo.github.io/>